

Article

The Role of Digital Technologies and Intellectual Property Management in Driving Sustainable Innovation

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Abstract: Innovative technologies have become a key driver of global development, enhancing communication, streamlining processes, and facilitating institutional collaboration. This study examines the perceptions of researchers and academic staff involved in research activities within educational institutions in Romania, a country that has faced challenges in intellectual property management and registration processes. The research focuses on the innovation registration process and the need for an online platform leveraging new digital technologies for intellectual property protection. Using an econometric model developed and tested through WarpPLS 8.0, the analysis reveals that 84% of respondents’ perceptions of the innovation registration process can be explained by the independent variables at the pattern level. The findings underscore the importance of aligning global legislative frameworks to stimulate innovation, promote intellectual property rights registration, and support sustainable economic growth. Furthermore, the study advocates for the development of an online platform that integrates artificial intelligence to enhance its functionality, streamline the application process, and foster the creation and dissemination of innovative technologies. By addressing these issues in Romania, the research contributes to global sustainable development objectives by providing a model for improving innovation registration processes worldwide, thus promoting technological advancement, intellectual property protection, and sustainable economic growth.

Keywords: innovative technologies; online platforms; digital transformation; technological innovations; intellectual property; technology transfer; digital society; globalization



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1. Introduction

Intellectual property (IP) represents a key concept established in the literature, often defined as the legal protection granted to original creations of the human mind. Historically, people have sought to protect their innovations, with references to intellectual property dating back to 500 BC, when patents were first granted for products made by individuals [1]. Over time, IP has become integral to research, development, and innovation activities [2], with intellectual property rights considered a direct result of innovation [3].

As defined by the World Intellectual Property Organization [4], IP encompasses a range of human creations [5], including inventions, literary and artistic works, and designs or symbols used in commerce [6,7]. These rights serve as legal safeguards for researchers, inventors, and creators [8], promoting innovation and ensuring the economic protection of their work [9] for a certain period of time [10]. IP laws not only protect the rights of creators but also play a crucial role in fostering global competitiveness and supporting economic growth [11,12]. Furthermore, IP catalyzes sustainable development [13], influencing sectors such as education, health, economy, and politics [14–16]. The role of digital technologies in IP management has grown increasingly significant in recent years. Innovations in digital platforms have made it easier to register and protect intellectual property, but challenges remain in ensuring these processes are harmonized internationally. Countries are increasingly adopting global standards and implementing measures to align their national IP laws with international frameworks, thus facilitating smoother research and development processes. These efforts are crucial in addressing barriers that hinder researchers' access to IP protection and in fostering international cooperation in innovation [17].

Over time, previous studies have highlighted the need for the harmonization of the innovation registration system as well as the existing legislation. Legislative harmonization plays a crucial role in facilitating the protection of intellectual property rights, and most importantly, in fostering innovation on a global scale. Currently, there is a strong emphasis within the European Union on harmonizing the existing legislation in the field of intellectual property protection. In this context, the European Patent Convention (EPC) and the Unitary Patent System aim to streamline the patenting process in member states, thereby reducing administrative barriers. Furthermore, their role is to ensure more effective protection of innovations. Directive 2004/48/EC on the Enforcement of Intellectual Property Rights provides a framework for the enforcement of intellectual property rights across all EU member states, addressing inconsistencies in national legal systems and strengthening cross-border cooperation.

From a global perspective, it should be noted that agreements such as the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) within the World Trade Organization (WTO) aim to align intellectual property legislation across countries, thus ensuring minimum standards for intellectual property protection worldwide. Moreover, the World Intellectual Property Organization (WIPO) facilitates international treaties such as the Patent Cooperation Treaty (PCT), which simplifies the process for patent applicants seeking protection in multiple countries.

It is evident that, both at the EU level and globally, there is significant concern regarding the harmonization of intellectual property legislation. These efforts aim to provide a more predictable legal environment that supports innovation. However, there are still challenges regarding the adoption of national legislation within these global frameworks. Further alignment of domestic laws with international standards is essential to create a cohesive global intellectual property framework.

Currently, the specialized literature includes numerous studies aimed at identifying systems for protecting intellectual property rights (IPR), as well as analyzing existing legislation in this field. A comparative study between developed (the UK) and developing countries (Turkey) revealed that, in nations with stronger legislative frameworks, the volume of foreign direct investment (FDI) is significantly higher, and the number of registered intellectual property rights is also considerably greater. Furthermore, it has been observed that developing countries [18] are increasingly motivated to improve their legislation and align it with international standards.

These studies also highlight the necessity of harmonizing intellectual property rights systems, with the ultimate goal of facilitating commercial relations between states [19].

Additionally, previous research indicates that developing countries exert significant effort to ensure that their innovations [20] are recognized. However, the legislation in these countries is often adopted from European or international models but proves difficult to implement effectively at the national level.

Another study has shown that small developing countries are compelled to adopt international legislative standards promoted by major advanced economies, such as the United States and the European Union, while large emerging economies are better positioned to resist such pressures. It has also been demonstrated that, while developed countries adopt strong policies for IPR protection, stricter IPR policies in poorer countries do not necessarily lead to increased innovation. In fact, strong but poorly implemented legislation in developing countries may hinder innovation [21]. Moreover, research indicates that merely strengthening IPR at the national level is insufficient to stimulate technology contracting in developing economies. Countries that provide stronger protection for IPR tend to experience positive effects on international knowledge contracts. More importantly, eliminating institutional and economic barriers can significantly enhance technology transfer and attract patent licensing agreements with foreign companies [22].

Another critical aspect that requires harmonization at the national level concerns the cost of registering intellectual property rights. Previous studies in this field have shown that the fees associated with applying for and registering an innovation in certain European countries are currently three to five times higher than those in Japan and the United States. This significant cost disparity often prevents individual researchers and small- and medium-sized enterprises (SMEs) from accessing the necessary financial resources, leading many to abandon the registration of their innovations altogether. Consequently, it is essential that, both at the European and global levels, similar and accessible cost structures are established for the registration and protection of intellectual property rights in order to foster innovation and ensure equitable access to protection mechanisms worldwide [23].

Based on the insights derived from prior research in this domain, there emerges a critical need for the harmonization of intellectual property registration systems alongside national legal frameworks. This alignment is pivotal not only for enhancing the efficiency of technology transfer but also for fostering sustainable development across countries.

At the international level, institutions like WIPO and the World Trade Organization (WTO) regulate IP, offering a global platform for cooperation and policy development [24]. Within the European Union, a network of institutions, including the European Commission, European Parliament, and European Council, oversee IP laws, while the European Union Intellectual Property Office (EUIPO) manages trademarks, designs, and models. Despite these efforts, one of the key challenges remains the consistent application of IP laws across Member States [25]. Recent legislative advancements in Romania reflect a growing effort to harmonize national laws with global IP standards, aligning with international agreements like the TRIPS Agreement and the Paris Convention. These reforms aim to streamline the registration process for innovations, remove existing barriers, and support Romania's integration into global research and innovation networks. However, obstacles still exist, particularly in ensuring adequate protection for researchers at the national level. Both global and local IP frameworks play a critical role in facilitating international collaboration, fostering technological transfer, and stimulating innovation [26].

In recent years, there has been a noticeable lack of satisfaction among researchers in Romania who are engaged in the development of innovations based on disruptive technologies, particularly regarding the process of registering these innovations. Many researchers tend to avoid registering their innovations due to the complexity of the process, as well as the absence of an online platform that would allow them to communicate with the relevant personnel, upload necessary documents, track updates, and identify the

next steps in the registration process. This lack of a digital solution has created significant barriers, increasing the time and effort required to navigate the system, further discouraging researchers from pursuing intellectual property protection for their innovations.

Consequently, they are forced to visit the relevant institutions in person to resolve these issues and submit their documents. This cumbersome and inefficient process not only hinders innovation but also discourages collaboration and progress in the fields of research and development. These challenges highlight the urgent need to create an online platform and simplify administrative procedures to foster a more supportive and streamlined environment for innovation.

1.1. Driving Transformation: The Crucial Role of Disruptive Technologies in Shaping Innovation and Sustainability

Novel digital technologies have the ability to transform existing business models as well as supply chains, thereby leading to disruptions in industries and markets [27]. The term “disruption”, according to the definition provided by the Cambridge dictionary, refers to preventing something, particularly a system, event, or process, from continuing as usual or as expected. Disruptive technologies possess the capacity to alter normal processes within society, industry, or companies at the local, national, and even global levels [28,29]. These technologies bring immersive benefits through dedicated instruments, improving the economy, the well-being of the population, and preserving the environment—three pillars of sustainability: economic, social, and environmental [30].

The term “disruptive” assigned to these technologies refers to their significant computational power, vast quantities of data, and unprecedented technological progress. They create transformative changes in society by reshaping economic sectors, labor markets, and consumption behaviors [31]. Disruptive technologies not only create new markets but also change existing products, services, and business models, potentially improving performance and reducing production costs. This innovation drives productivity and efficiency, offering significant opportunities to industries and businesses. Their disruptive nature can replace outdated products or services with new or optimized alternatives [32]. Industry 4.0 represents a technological collaboration aimed at transforming traditional processes through digitalization [33]. Digitalization integrates digital technologies into business processes with the goal of achieving improvements at both the functional and organizational levels. These disruptive technologies significantly advance industries’ operations. Digital platforms, as a novel approach, alter organizational processes by introducing a business ecosystem that changes value networks [34].

Digital technologies have radically transformed how businesses operate and how societies function, reshaping economic dynamics [35]. Blockchain, an emerging disruptive technology, facilitates peer-to-peer value transfers, ranging from digital currencies to physical commodities, without relying on intermediaries. This technology enhances efficiency, transparency, and security within business transactions, ensuring industry transparency and streamlining supply chains [36]. Blockchain is also changing the way people interact by leveraging technologies like smart contracts, which eliminate the need for trust between individuals or organizations through intermediaries. These contracts are executed automatically via technical code. Blockchain is further enabling social innovations such as electronic voting, degree verification, and land registries. Moreover, blockchain plays a significant role in healthcare platforms by enhancing security, performance, data integrity, and privacy. It can also host other technologies, including artificial intelligence (AI) and the Internet of Things (IoT) [37]. AI is a central technology within Industry 4.0, aiming to replicate human intelligence in machines, such as robots or software. AI has demonstrated impressive growth over recent years and continues to develop, particularly in applications that impact economic, social, and environmental spheres. It has a significant role in healthcare, where

it helps diagnose human conditions by recognizing patterns from training data and aids medical professionals in creating personalized treatment plans [38,39].

IoT, another key technology in Industry 4.0, connects devices and applications via the Internet, establishing an integration between the physical world and cyberspace. IoT platforms have facilitated significant advancements in the industrial sector by enabling devices to communicate via the Internet [40]. Both IoT and blockchain technologies share an important feature: scalability, which plays a critical role in intelligent network communication, allowing for real-time data exchange between devices [41]. Looking ahead, Industry 5.0 will build on the foundation of Industry 4.0, with a more human-centered approach that prioritizes resilience and sustainability. Society 5.0 envisions a highly integrated cyber-physical platform designed to support sustainable development goals, such as generating prosperity, ending poverty, and protecting the Earth [42].

A previous study has illustrated the strong connection between innovation and institutional quality in the adoption and utilization of new technologies, aimed at responding to rapid changes in the external environment and supporting countries in achieving sustainable development across various sectors [43]. In light of the findings obtained in this study, it is important to note that the development of user-friendly digital tools for registering and managing intellectual property can significantly enhance the diffusion of sustainable technologies. By simplifying access to innovation protection and ensuring transparent, efficient, and accessible processes through digital platforms, such tools can address barriers related to institutional inefficiencies and affordability. This, in turn, can encourage greater adoption of green technologies at both the household and industry levels. Therefore, integrating digital technologies into the innovation ecosystem not only supports effective intellectual property management but also directly contributes to increasing the adoption rates of sustainable technologies.

Recent research underscores the critical role of technological innovation in mitigating greenhouse gas emissions, while highlighting the need for sector-specific fiscal instruments, such as carbon taxes, to enhance policy effectiveness. In response to environmental challenges, many OECD countries have shifted from reactive measures to proactive strategies, prioritizing investments in sustainable technologies and implementing comprehensive regulatory frameworks, including carbon pricing schemes like taxes and tradable permits. These mechanisms not only stimulate the adoption of green innovations but also provide financial resources to support their development, in alignment with international climate commitments such as the Paris Agreement and the Sustainable Development Goals. In this context, digital technologies are pivotal in driving and scaling sustainable innovation, while efficient intellectual property management ensures legal protection, promotes commercialization, and encourages continued investment. The integration of advanced digital platforms with robust intellectual property frameworks is therefore fundamental to building an innovation-driven, sustainable economy [44].

Another study conducted in this field has highlighted the importance of technological innovation in environmental protection for achieving the Sustainable Development Goals (SDGs) in European Union countries. The analysis revealed the existence of a bidirectional relationship between environmental technological innovation (ETI) and the SDGs, emphasizing that, in certain countries, such as Germany, the Netherlands, and Spain, SDG-oriented policies drive technological progress, while, in others, including Austria, France, and Sweden, technological innovations significantly influence the attainment of SDGs. The findings of this research illustrated the need to integrate sustainable development principles into national innovation strategies through the establishment of dedicated programs, targeted funding, and favorable regulatory frameworks. In light of these findings, it becomes evident that digital technologies, combined with effective intellectual property manage-

ment, can serve as key drivers in fostering sustainable innovation, contributing to the alignment of technological progress with global sustainable development objectives [45].

Industry 5.0 represents a revolutionary shift where human and machine collaboration enhances production efficiency. This system incorporates advanced manufacturing and automation, with the human element at its core. It will utilize platforms like the IoT and the Internet of Services (IoS), as well as software, robots, and other cutting-edge technologies to facilitate connectivity and data exchange [46].

Industry 5.0 will combine human intelligence with cognitive computing to create value-added products, leveraging the best of industrial automation and critical thinking. This new system is characterized by real-time data availability and high connectivity throughout the entire ecosystem [47] (Singh et al., 2023). The intellectual property (IP) rights system was developed to reward innovation through monopoly rents. However, recent concerns have arisen regarding the negative effects of excessive IP protection, particularly in developing countries. The international IP system must ensure that technological advancements are widely distributed to avoid creating monopolies and restricting access [48]. In collaborative environments, intellectual property plays a crucial role, as sharing proprietary knowledge and technological insights is unavoidable. Disruptive technologies often embody cutting-edge innovations, offering significant competitive advantages. If an organization loses control over its IP, it may diminish its ability to fully capitalize on technological advancements [49]. The rise of AI further complicates the IP landscape due to its potential for continuous knowledge evolution. Algorithms, although privately kept, must be disclosed for transparency, especially to determine if they exhibit discriminatory behavior [50]. The effectiveness of open innovation in disruptive technologies requires a balance between IP protection and collaboration. Organizations must adopt strategies like selective disclosure and structured agreements to safeguard intellectual property rights while benefiting from open innovation [51].

1.2. The Impact of Disruptive Technologies on the Evolution of Digital Public Platforms for Sustainable Governance

The rapid expansion of digitalization and Information and Communication Technologies (ICT) has significantly impacted public service delivery, particularly in the 21st century. These advancements promise to enhance the accessibility, efficiency, and effectiveness of public services [52]. ICT facilitates the flow of information between governments, citizens, and institutions, thereby streamlining communication and fostering transparency. The Internet, as the most crucial digital medium, plays a key role in this process [53]. Digital platforms, in particular, have disrupted not only markets but also civic and social practices, fundamentally altering public communication and democratic processes [54]. The dominance of global corporations over the online infrastructure for news, video, and private communication has raised concerns regarding the potential compromise of public values such as democracy, transparency, and tolerance.

This phenomenon, often referred to as the process of platformization, has not only transformed markets but also reshaped the organizational and infrastructural designs of societies [55]. As a result, governments have increasingly adopted digital services to enhance their capabilities in serving citizens. Digital services, as part of electronic government, rely on ICT tools such as web-enabled devices or electronic data management systems to automate processes and simplify bureaucratic procedures [56]. One notable tool integrated into these digital platforms is neural networks. These AI-powered learning algorithms analyze large volumes of data to identify patterns that inform better policymaking and improve the implementation of public initiatives [57]. Moreover, digital public platforms have been instrumental in reducing corrupt behaviors by enabling more effective oversight of public officials. Through smart city initiatives, these platforms foster enhanced

stakeholder engagement and prioritize citizens in the governance process. The platforms facilitate technologically mediated interactions between user groups, supporting public participation and collaboration [58].

Digital platforms exhibit various economic, business, and governance dimensions, all contributing to the creation and capture of value [59]. In the economic domain, these platforms operate through the dynamics of demand and supply within market environments. From an industrial perspective, digital platforms are shaped by market capitalization and governance models. They also represent a socio-technical phenomenon, influencing development in unique ways [60]. In innovation platforms, governance plays a crucial role in determining who can participate and contribute to the platform's evolution. Effective governance ensures sustainable benefits for all stakeholders, including users and complementors. Examples of governance mechanisms include regulating access to the marketplace, establishing grading systems, and controlling access to platform-generated information [61].

A variety of digital technologies, such as blockchain, artificial intelligence (AI), and the Internet of Things (IoT), can be employed within a digital platform for innovation registration. Blockchain technology can be implemented to ensure transparency, data security, and traceability. This technology enables the storage of innovation-related data in a decentralized ledger, ensuring the protection of intellectual property rights under the most secure conditions. Moreover, through its use, each submitted innovation can receive a timestamp at the moment of registration, allowing for the precise identification of the submission date. Blockchain can also be utilized to automate the processes of innovation evaluation and validation, as well as to facilitate the payment of fees associated with the registration process.

The integration of AI into this process can bring a series of benefits. For instance, AI can provide virtual assistance through the implementation of a chatbot, offering real-time guidance to users. Furthermore, AI enables the automatic analysis of documents to detect potential similarities with previously submitted innovations. Additionally, this technology allows for the generation of predictions and tailored recommendations to support the innovation process.

The use of the IoT within a digital platform for innovation registration offers notable advantages, particularly in cases where inventors are registering physical prototypes. For such innovations, the IoT can be employed to transmit real-time data regarding the functionality of the prototypes, thereby facilitating the evaluation process. Furthermore, the IoT can significantly contribute to the monitoring of patent usage, providing valuable insights into the practical application of certain innovations across various industries. This enables a comprehensive assessment of the necessity for updating or extending specific intellectual property rights.

In the context of the rapidly evolving digital and technological landscape, the research conducted in this paper aims to explore the perceptions of researchers and academic staff engaged in research activities regarding the process of registering innovations for the protection of intellectual property rights in Romania. The primary focus is on the need to create an online platform that facilitates the innovation registration process, exploring how digital technologies, such as artificial intelligence and automation, can be leveraged to streamline operations, enhance the user experience, and improve the overall efficiency and reliability of the platform.

The use of artificial intelligence within digital platforms enhances the efficiency of innovation protection processes, reduces both the time and cost associated with registering innovations, and streamlines communication between researchers and authorized personnel within institutions. All of these aspects ultimately stimulate innovation and contribute

substantially to the development of a modern, accessible, and efficient digital infrastructure, directly aligning with the objectives of SDG 9. Furthermore, the implementation of AI in such digital platforms facilitates collaboration between national and international institutions, as well as among researchers, inventors, and investors. These platforms therefore support the strategic partnerships necessary for the implementation and promotion of innovations on a global scale, which aligns with SDG 17. In light of these considerations, it must be emphasized that the integration of artificial intelligence into digital platforms for innovation registration has the potential to advance both technological processes and international cooperation, all aimed at achieving global sustainable development goals.

Given the increasing significance of disruptive technologies and their far-reaching implications for global innovation, this study aims to critically evaluate the optimization of the innovation registration process and its alignment with international standards through the integration of emerging technological advancements. Such alignment is paramount not only for ensuring the robust protection of intellectual property rights but also for fostering a more efficient and collaborative ecosystem for research, development, and innovation on a global scale.

The findings of this research seek to provide a comprehensive understanding of how the registration process can be structured to incentivize researchers to formally register their innovations with accredited institutions. Moreover, the study will generate valuable insights into the legislative harmonization required within the framework of an online platform dedicated to intellectual property protection. Ultimately, the results will underscore the pivotal role of emerging technologies in enhancing the efficiency and effectiveness of intellectual property rights registration processes.

1.3. Research Objectives

The study aims to achieve the following objectives:

OB1: To evaluate the influence of perceived facilities during the innovation registration process on respondents' overall perception.

OB2: To analyze the effect of problems encountered by respondents during the registration process on their perception of its effectiveness.

OB3: To identify how the time and effort invested by researchers in the registration process influence their perception of its efficiency.

OB4: To determine the role of respondents' prior knowledge and experience in shaping their perception of the registration process.

OB5: To investigate the perceived risks and their effect on respondents' perception.

OB6: To examine the influence of perceived registration costs on respondents' overall perception of the process.

OB7: To explore the relationship between respondents' perception of the registration process and their intention to register future innovations.

1.4. Research Hypotheses

In order to achieve the objectives shown above, the study tests the following hypotheses:

H1. *The facilities perceived by the respondents regarding the innovation registration (for the protection of the intellectual property rights) positively influence their perception of the innovation registration process. This aspect is supported by the IS Success model [62] and by the SERVQUAL model [63].*

H2. *The problems perceived by the respondents negatively influence their perception of the process, as highlighted by the SERVQUAL model [64].*

H3. *The time and effort invested by researchers in the innovation registration process for the protection of the intellectual property rights negatively influence their perception of this process, as confirmed by the results of Baker [65].*

H4. *The respondents' knowledge and experience regarding the steps to be taken during this process positively influence their perception of it, similar to the conclusions of the SERVQUAL model [66].*

H5. *The risks perceived by the researchers regarding the innovation registration process (for the protection of the intellectual property rights) negatively influence their perception of this process, a phenomenon previously reviewed by the SERVQUAL model.*

H6. *The costs perceived during this process negatively influence the respondents' perception, as emphasized by Dodds, Monroe, and Grewal [67], and Agarwal and Teas [68].*

H7. *The respondents' perception regarding the innovation registration process for the protection of the intellectual property rights positively influences their intention to register future innovations, as confirmed by the SERVQUAL model.*

The results generated by this study will highlight the barriers and opportunities encountered during the innovation registration process in Romania, particularly in light of the ongoing digital transformation. These insights will not only shed light on the challenges faced by researchers in securing intellectual property rights but also offer recommendations for improving the registration process, enhancing the user experience through digital innovation, and harmonizing the process with international standards. Ultimately, the study aims to contribute to strengthening the protection of intellectual property in a digital era, fostering a more collaborative and efficient research, development, and innovation ecosystem.

A conceptual model (Figure 1) has been developed based on the hypotheses outlined above and will be tested throughout this study to identify the critical factors influencing researchers' perceptions and behavior toward the digital registration of innovations for intellectual property protection.

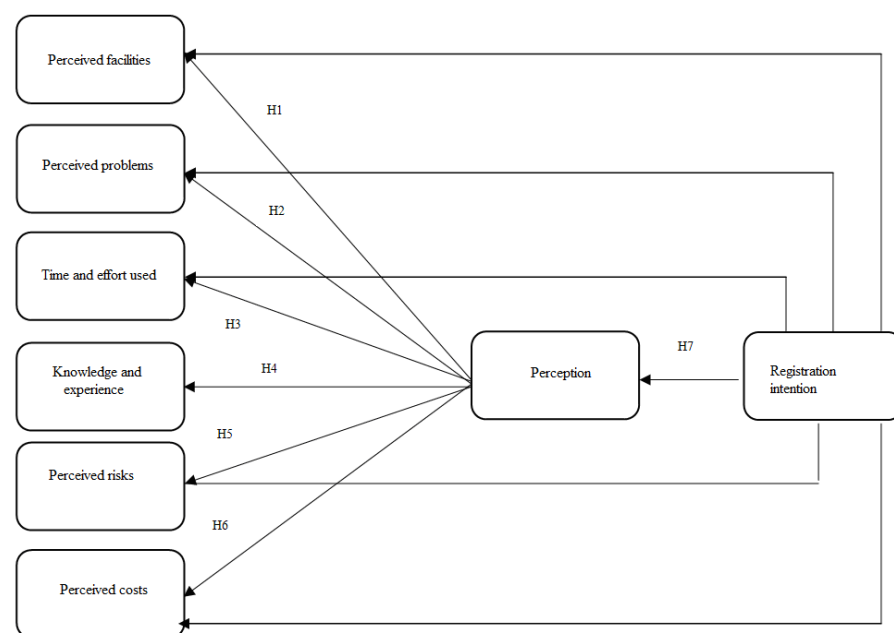


Figure 1. Proposed conceptual model.

2. Materials and Methods

The study shown in this paper was conducted between 1st September 2023 and 10th January 2024, thus ensuring the relevance and timeliness of the results obtained, with the said results being shown in the Doctoral Thesis “Technological Transfer and Intellectual Property in Romania” [69]. The primary objective of this study was to capture the current perceptions and experiences of researchers and academic staff in Romania regarding the process of registering innovations for the protection of intellectual property rights. A key focus was placed on identifying the need for the creation of an online platform to facilitate this process, particularly in the context of digital transformation and the increasing reliance on online platforms. For the collection of information, a structured questionnaire was developed and distributed through an online platform. The potential respondents were contacted by e-mail and were invited to fill out the questionnaire by accessing a specific link. They were also asked to forward the questionnaire to their colleagues in the research teams, thus contributing to a broader perspective regarding the topic reviewed.

The study targeted the teaching staff and researchers working in educational institutions in Romania regardless of their status (teacher, researcher, or other roles associated with the research activities). The final sample of the research included 244 respondents, thus providing a solid basis for the analysis of the perceptions and for the identification of the determining factors during the innovation registration process for the protection of the intellectual property rights.

The approach used in this study, i.e., the collection of data online and the mobilization of the participants through the research teams, allowed for a wide dissemination of the questionnaire and a diversity of respondents. The results thus obtained provide a comprehensive and relevant perspective regarding the topic under review, therefore contributing to the development of a framework for understanding the challenges and opportunities encountered during the protection of the intellectual property rights in Romania.

Questionnaire Development

Pursuant to the objectives and hypotheses shown above, the questionnaire used in this study was developed in order to ensure the relevance and validity of the data collected. In order to filter only respondents belonging to the research population, a filter question was included and became the starting point of the selection process.

The questionnaire was divided into the following four main sections:

- Filter question: Q1—meant to identify the individuals eligible for participation;
- Introductory questions: Q2–Q18—meant to provide an overall view of the professional activity carried out by the respondents;
- Questions reflecting the variables of the conceptual model: perceived facilities (Q19), perceived problems (Q20), time and effort used (Q21), knowledge and experience during the innovation registration process (Q22), perceived risks (Q23), perceived costs (Q24), overall perception regarding the registration process for the protection of the intellectual property rights (Q25), and the intention to register an innovation in the future (Q26).
- Demographic questions: Q27–Q31—meant to record variables such as respondents' gender, age, education background, income, and residential environment.

The 7-point Likert scale, with intervals between 1 (Strongly Disagree) and 7 (Strongly Agree), was used for the measurement of the variables included in the conceptual model, thus ensuring an adequate granularity in capturing the perceptions. Moreover, the nominal and interval scales were used for the relevant questions depending on the nature of the data concerned. In order to ensure the clarity and relevance of the questionnaire, it was pre-tested on a sample of 10 respondents. Following the pre-testing, the issues related to the

development of the questions, the coherence of the writing, and the clarity of the expression were identified and corrected. After the remedial of all of these aspects, the final version of the questionnaire was distributed via an online platform. This stage of development and testing of the questionnaire enabled the creation of a robust research instrument capable of accurately capturing the variables outlined in the conceptual model. It contributed to a deeper understanding of the respondents' perceptions regarding the innovation registration process and the necessity of creating an online platform in Romania, particularly in the context of the ongoing digital transformation and the role of technological advancements in enhancing the efficiency and user experience of intellectual property protection systems.

From the perspective of how the questions were structured, it should be noted that, for each variable, a series of statements were presented, and the respondents were required to express their agreement or disagreement with these statements. Regarding the facilities obtained through the innovation registration process within an authorized institution, the statements included information such as the legal protection provided for innovations after registration, the potential to secure additional resources, access to European funding, and the enhancement of the inventor's image or status, among others. Concerning the main issues addressed, the statements covered details such as the large number of documents required during this process, the lack of a standardized model for completing certain documents, the numerous stages that need to be followed, the long response times to requests, and the absence of an online platform to facilitate the process.

With respect to the time and effort required for innovation registration, the statements in this category included details about the lengthy time needed for innovation development, the time required to obtain information about the steps involved in registration, and the significant time needed to compile the necessary documentation for submission to an authorized institution. Regarding the experience and knowledge of respondents related to the innovation registration process, researchers were asked whether they agreed or disagreed with statements such as having previously submitted a request for the registration of another innovation, having participated in preparing a file for the registration of an innovation, or having previously registered another innovation within an authorized institution.

In terms of perceived risks, respondents' agreement or disagreement was assessed on issues such as the possibility that employees at authorized institutions may not consider the researchers' proposal as innovative, the potential similarity of the innovation to others, leading to the rejection of the application, the chance that the reviewers may deem the innovation uninteresting, or the possibility that the information submitted might be disclosed. Finally, analyzing from the perspective of perceived costs, the statements for this question focused on the costs associated with developing an innovation and the costs involved in the innovation registration process.

3. Results

The dissemination of the questionnaire revealed that 244 of the respondents carried out innovation activities or were members of a research team that aimed to innovate in a specific field over the last 5 years.

3.1. Respondent Profile

A total of 63.1% of respondents are teachers who are currently carrying out research activities, while 36.9% are researchers. A total of 36.1% of them work in a higher education institution, 25% of them work in a high school, 36.9% of them work in a research center, and 2% of them work in a cultural center. The fields in which the respondents generally carry out research activities are the following: humanities (15.6% of those who participated in the

study), health (13.9% of the respondents), economic sciences (12.5% of the respondents), physics (11.8% of the respondents), applied life sciences and biotechnology (10.1% of those who participated in the study), engineering sciences (8.3% of the respondents), sciences of materials (3.5% of the respondents), chemistry (4.2% of the respondents), social sciences (3.5% of the respondents), mathematics (2.4% of the respondents), biology and ecology (2.4% of the respondents), and earth sciences (1.7% of those who participated in the study). They carry out research activities most often in mathematics, computer science, chemistry, physics, and engineering sciences. In terms of the number of innovative projects in which the respondents had participated so far, 49.2% of the respondents had worked on only one innovative research project, 35.2% of them had taken part in two to three projects, 7.8% of those interviewed stated that they had participated in more than six such projects so far, and 7.8% of them had participated in four to six projects.

The review of the respondents in terms of demographic variables taken into account at the quantitative study level revealed that 61.5% of them were women and 38.5% were men. A total of 80% of them considered that the technology transfer is of increased significance for the country. In terms of the age range of those who participated in the study, it should be noted that 17.6% of them were between 30 and 39 years old, 48% of them were between 40 and 49 years old, 16.8% of the respondents were between 50 and 59 years old, while 3.3% of the individuals were between 60 and 69 years old. In terms of the last education institution graduated by the respondents, 18% of them had completed their doctoral studies, 66.4% of them had completed their Master's studies, while 15.6% of the individuals had completed their bachelor's studies. In terms of the net income earned monthly by the respondents, 25.4% of them had a net income of over RON 7000 per month, 17.2% of them received a monthly income between RON 6001 and 7000, 49.6% of them earned between RON 5001 and 6000, and 7.8% of them had a net monthly income of RON 4001–5000. In terms of the respondents' residential environment, 93.9% of them lived in urban areas, while 6.1% of them lived in rural areas.

3.2. Respondents' Perception Regarding the Intellectual Property Protection Process

For the review of the respondents in terms of people who registered their innovations with an institution authorized to protect their intellectual property rights, it should be noted that 36.9% of those who participated in the study did so, while 63.1% had not registered their innovations up to that point. Thus, the respondents' perception regarding the intellectual property protection process was conducted exclusively on individuals who registered at least one innovation to date with a specialized and authorized institution (total: 90 respondents). The main reason that led the respondents to register their innovations with an authorized institution was the desire to gain financial resources (e.g., for commercial purposes), at 33%, while 33.3% of them wanted to have access to a project financed by European funds, and only 27.8% did so out of the desire to protect their intellectual property. A total of 5.6% of those who participated in the study did so for prestige.

Concerning the review of the respondents in terms of the information sources used by the future researchers in order to document themselves regarding the steps they must take to protect their intellectual property with the help of an authorized institution, 44.4% of them stated that they had gathered their information from the website of that institution, 23.3% of them had read the information available on various specialized forums in this field, 16.7% of them had studied the information available on the website of the educational institution/cultural center/research center they belonged to, 10% of them had discussed with their friends/colleagues who had previously gone through this process, and 5.6% of the respondents had gathered their information from social media (from the specialized

groups existing at their level). In terms of the difficulties encountered during the innovation registration process for the protection of the intellectual property, the results reveal that 38.9% of the respondents mentioned that they had some impediments at this level, while 61.1% of them stated that they had not encountered such problems. The main difficulty encountered by the respondents during the innovation registration within an authorized institution was the complex process (the high number of documents requested) (42.9%). A total of 28.6% of them considered that the lack of information regarding the stages that must be completed in order to protect the intellectual property represents the main impediment, while 28.6% of them considered that the long time required for the review of the submitted files was the main problem during this process. A total of 85.7% of those who participated in this study believed that the innovation registration process should be improved in the future in order to protect the intellectual property, while 14.3% believed that this was not necessary, with the current process being efficient. Moreover, the need for harmony between the intellectual property protection process at national level and at European level, and also at global level, was observed.

In terms of the way that the innovation registration process should be improved in order to protect the intellectual property, 33.3% of those who participated in the study mentioned that an online platform should be created for the submission of the necessary documents for registering the innovation and for automatically receiving a registration number, 33.3% of the individuals believed that the number of documents requested in the process should be reduced, 16.7% of those interviewed considered that an online platform should be created for the receipt and transmission of the documents during the innovation registration process, while 16.7% of them considered that it was necessary to reduce the stages that must be completed during the process. Thus, there should be a standard process both at global and European level, at which the common laws that must be complied with, the stages that must be completed, the necessary documents, and the additional information tools they can use to document themselves regarding this process should be listed.

From the perspective of the key advantages that such a platform would offer to researchers wishing to register their innovations, 35% of them indicated that the platform would facilitate the submission of documents to the relevant authorities, 28% of researchers considered that the platform would enhance communication with institutional staff, 15% mentioned that the platform would assist in quickly identifying potential issues with the innovations, and 12% highlighted that the platform would reduce the time required for the registration process within the authorized institution. A total of 10% of participants noted that it would help lower the registration costs for innovations within the institution.

From the perspective of the main challenges that could arise in the implementation of such a platform, 30% of respondents mentioned that one of the key issues is the lack of a robust and reliable technological infrastructure, 22% indicated the poor coordination between the main state institutions, 18% highlighted the resistance to change among employees within the authorized institution, and 15% referred to issues related to the existing legislation. A total of 9% pointed out the financial constraints within the state budget for developing such an innovation registration platform. A total of 6% mentioned potential concerns regarding data confidentiality and security. A significant majority (80%) of the study participants considered the integration of artificial intelligence within the online platform to be of considerable importance. Among the respondents, 23% suggested that AI should be leveraged to automate the verification of documents submitted by researchers, 21% recommended its use in developing an AI-powered interface, such as a virtual assistant, 20% emphasized the potential of AI in identifying plagiarism and similar innovations, 15% of participants proposed the application of AI to offer intelligent suggestions for completing

submissions, while 13% advocated for its role in automating the evaluation process of the submitted applications. Lastly, 8% of respondents indicated that AI could be used to optimize the overall workflow of the platform.

Regarding the respondents' perception of the extent to which the legislative aspects currently existing at the national level have the capacity to protect the individuals' intellectual property, they considered that they had the capacity to protect them to a certain extent acceptable, the average calculated being 3.33. For this particular reason, in the future, there is a need to harmonize the national legislative measures with the global measures so that the respondents' perception regarding this aspect is as good as possible, and so that they feel safe from this point of view.

3.3. SEM Analysis Using WarpPLS 8.0

The method chosen for testing the proposed research model was the variation method. WarpPLS 8.0 software was used for its testing (Figure 2). This is a software specialized for structural equation modeling (SEM) that uses the partial least squares (PLS) method.

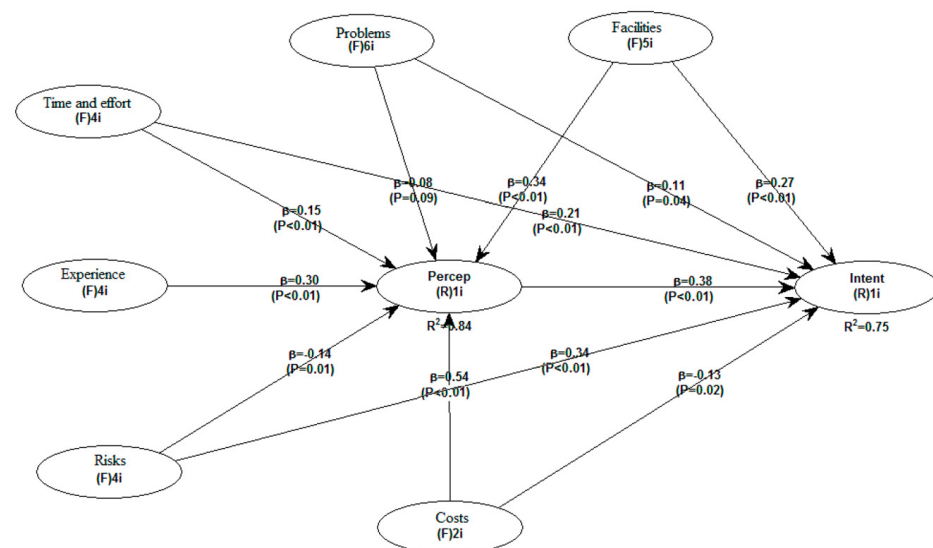


Figure 2. The model of the perception of researchers and academic staff engaged in research activities regarding the process of registering innovations for the protection of intellectual property rights in Romania.

The SEM method using WarpPLS 8.0 was selected due to its robustness in analyzing the intricate relationships between the variables involved in the innovation registration process. This methodology allows for a nuanced evaluation of the reciprocal influences among the identified factors, thereby providing a thorough understanding of the perceptions held by researchers and academic staff regarding intellectual property protection processes. Additionally, SEM facilitates the testing and validation of complex theoretical models, ensuring the derivation of robust and reliable results that align with the research objectives.

The analysis conducted for this research aimed to explore how a range of variables influence the perceptions of both future researchers within educational institutions and researchers within research centers concerning the innovation registration process for intellectual property rights protection. The latent variables incorporated into the model included the perceived facilities, perceived challenges, time and effort expended, future researchers' knowledge and experience, perceived risks, and perceived costs.

The hypotheses of this research had the role of identifying the existing connection between several variables. In the table below, the hypotheses initially established, as well as the Beta coefficients and the significance thresholds, can be observed.

Following the testing of the model, as can be seen in Table 1, six hypotheses were accepted out of the total of seven initially established for the proposed conceptual model. Hypothesis 2 was not accepted due to the results obtained from the analysis in WarpPLS. Specifically, the Beta value for this hypothesis was found to be 0.08, which is significantly lower than the threshold value of 1. Additionally, the significance level for Hypothesis 2 exceeded the conventional threshold of 0.05, with a p -value of 0.09. A p -value greater than 0.05 indicates that the observed relationship is not statistically significant, meaning there is insufficient evidence to support the hypothesis. Therefore, based on these criteria, Hypothesis 2 was rejected, as it did not meet the necessary statistical thresholds to be considered valid within the context of the analysis.

Table 1. The validation of the hypotheses using the variation method.

No.	Hypothesis	β	p	Validation
H1	Perceived facilities—Perception	0.34	<0.01	YES
H2	Perceived problems—Perception	0.08	0.09	NO
H3	Time and effort used—Perception	0.15	<0.01	YES
H4	Knowledge and experience—Perception	0.3	<0.01	YES
H5	Perceived risks—Perception	0.14	0.01	YES
H6	Perceived costs—Perception	0.54	<0.01	YES
H7	Perception—Intention	0.38	<0.01	YES

The review performed also studied the value of R^2 . This indicator illustrates the extent to which the independent variables of the model have the ability to explain, to a certain extent, the variation in the dependent variables. The review and interpretation of the results revealed that the values obtained for this indicator are quite high, which illustrates the independent variables taken into account at its level and which have the ability to largely explain the variation in the dependent variables. In conclusion, the future researchers' perception regarding the innovation registration process (with the aim of protecting the intellectual property) is 84% explained by the independent variables, and the researchers' intention to register their innovations is 75% explained by the independent variables taken into account by the model (Table 2).

Table 2. R^2 determination coefficients.

Respondents' Perception	Respondents' Intention
0.84	0.75

Several conformity indicators were used for testing the validity of the model. The results obtained show that the model is valid because the p -value was less than 0.001 for the APC, ARS, and AARS indicators. The AVIF value was $3.543 > 3$, which illustrates the fact that this model is accepted. The value of AFVIF was $3.082 < 3.3$, illustrating a very good result for the model, while the value of the GoF was 0.836.

The value of the SPR indicator was 0.833, a value higher than 0.7; the value of the RSCR was 0.908, a value higher than 0.9, and the value of the SSR and NLBCDR indicators was 1, a value higher than 0.7 (Table 3). Considering the values generated by all of the indicators obtained from this study, it can be stated that the proposed model is accepted.

Table 3. The coefficients used for measuring the conformity of the research model.

Indicators	Validation Criteria
Average path coefficient (APC) = 0.249	$p < 0.001$
Average R-squared (ARS) = 0.798	$p < 0.001$
Average adjusted R-squared (AARS) = 0.793	$p < 0.001$
Average block VIF (AVIF) = 3.543	Accepted if the value generated is ≤ 5 ; Ideal ≤ 3.3
Average full collinearity VIF (AFVIF) = 3.082	Accepted if the value generated is ≤ 5 ; Ideal ≤ 3.3
TenenhausGoF (GoF) = 0.836	Accepted if the value generated is low ≥ 0.1 , average ≥ 0.25 , high ≥ 0.36
Sympson's paradox ratio (SPR) = 0.833	Accepted if the value generated is ≥ 0.7 ; Ideal = 1
R-squared contribution ratio (RSCR) = 0.908	Accepted if the value generated is ≥ 0.9 ; Ideal = 1
Statistical suppression ratio (SSR) = 1.000	Accepted if the value generated is ≥ 0.7
Nonlinear bivariate causality direction ratio (NLBCDR) = 1.000	Accepted if the value generated is ≥ 0.7

4. Discussion

In the context of an increasingly interconnected global economy, intellectual property (IP) has become a crucial driver of innovation [70,71], foreign investment, and the protection of creators and inventors across all sectors [72]. Universities and research centers, as key contributors to scientific progress, are at the forefront of developing new solutions using disruptive technologies, which can then be registered and protected by the relevant institutions. Researchers in these institutions not only advance their respective fields but also contribute to the broader economic and social development [73]. This dynamic underscores the importance of intellectual property rights as a fundamental tool for enhancing global competitiveness and fostering sustainable economic growth.

Intellectual property rights provide researchers and innovators with exclusive legal rights over their creations, thus safeguarding their work against theft or imitation [74]. However, the protection of these rights remains complex due to the lack of harmonization between national IP laws, which can create significant challenges for international collaboration and the recognition of IP across jurisdictions. These discrepancies highlight the need for global standardization to simplify the IP registration process, reduce administrative burdens, and foster international cooperation [75].

Recent technological advancements [76], such as artificial intelligence, blockchain, big data, and the Internet of Things [77], have posed new challenges and opportunities in the IP domain [78–80]. The rapid growth of disruptive technologies has intensified the need for updated IP protection frameworks that can accommodate these emerging innovations, both online and in traditional sectors like healthcare, pharmaceuticals, and the environment [81,82]. These developments underscore the urgency of reforming national and international IP laws to ensure the effective protection of intellectual property rights in a rapidly evolving technological landscape [83].

The research presented in this paper aims to explore the perceptions of researchers and academic staff engaged in research activities regarding the process of registering innovations for the protection of intellectual property rights in Romania. Specifically, it focuses on understanding how digital technologies can be leveraged to optimize this registration process, improve efficiency, and streamline interactions with relevant institutions. Conducted with 244 respondents, the study revealed that 63.1% of respondents had not yet registered their innovations for IP protection, citing reasons such as complex procedures, lack of information, and delays in processing applications. Conversely, 36.9% had regis-

tered their innovations, often driven by the desire for financial resources, European funding opportunities, or prestige.

The respondents also highlighted significant challenges in the current registration process, including the cumbersome number of required documents and the slow pace of approval. The majority of participants advocated for improvements, with many suggesting the creation of a streamlined online platform to facilitate document submission and reduce bureaucratic inefficiencies. Furthermore, there was a strong consensus on the need to align Romania's IP registration process with international standards to ensure optimal protection for researchers' innovations.

The limitations of this study include the relatively small sample size of 244 respondents. The sample is considered adequate for the study conducted in this work, providing valuable insights into the analyzed topic. However, there are limitations regarding the generalization of the results. The findings derived from this sample may reflect the specific experiences and perspectives of the respondents within this study and, therefore, cannot be fully extended to a broader population. To overcome this limitation, future research could aim to expand the study by increasing the sample size, which would contribute to enhancing the robustness and reliability of the findings. A larger and more diverse sample would provide a more representative view of the innovation registration process and the challenges associated with it. Furthermore, it is deemed necessary to extend the study to include other countries, both within the European Union and globally. This would allow researchers to conduct comparative studies and identify key factors that differentiate the innovation registration process across these countries. Additionally, such future studies would aid in understanding how both the innovation registration process and existing legislation should be harmonized to promote sustainable development. By incorporating these future research directions, the external validity of the results would be enhanced, and a more comprehensive, global understanding of the factors influencing innovation registration and intellectual property protection could be achieved.

Additionally, the online survey distribution method may have led to a biased sample, as some potential respondents may not have had access to the survey link. Future studies should aim to reach a larger, more representative sample and explore qualitative methods, such as interviews or focus groups, to gain deeper insights into the challenges faced by researchers.

From the perspective of the need to protect innovations based on disruptive technologies, it is important to note that ensuring intellectual property protection in this field is particularly challenging. This is primarily due to the rapid pace at which these technologies become obsolete, often within a year, thus diminishing their relevance and economic value to society [84]. A major challenge in protecting intellectual property rights in the context of disruptive technologies lies in the need to adapt patent legislation to align with the limited timeframe [85] during which such innovations are considered novel and commercially viable. While the standard duration of patent protection is 20 years, this period is often ineffective for digital innovations, where the novelty and utility of the invention may decline significantly within six months to a year. Consequently, there is a pressing need to rethink and reform the legal framework governing intellectual property rights [86], both at the national and international levels, in accordance with the accelerated pace of technological progress.

Moreover, it is essential to identify which types of intellectual property rights are most appropriate for the protection of disruptive technologies [87]. Copyright law offers flexible protection and is particularly suitable for safeguarding software solutions and digital content derived from such technologies. On the other hand, while patents can provide strong legal protection, the current patenting process is often cumbersome and

time-consuming, potentially hindering innovation in rapidly evolving technological sectors. Nevertheless, if streamlined and modernized, the patent system could still play a crucial role in protecting innovations based on disruptive technologies.

In addition, utility models may serve as an intermediate form of protection. They typically offer faster, albeit shorter-term, protection and may be better suited to certain categories of digital innovations. This flexibility makes utility models a viable alternative for innovators seeking timely protection in fast-moving markets.

At the European level, there is an ongoing effort to address the challenges of protecting intellectual property in the context of disruptive technologies. The European Union's Digital Transformation Policy, including the Digital Agenda and the Strategies on Data and Artificial Intelligence, aims not only to foster the adoption of innovative digital technologies but also to adapt the regulatory framework to support and encourage innovation. AI has a key role in the creation, management, and dissemination of innovations. While AI can significantly enhance the efficiency of intellectual property registration processes, it also raises critical issues regarding eligibility for protection and the attribution of rights in cases where innovations are autonomously generated by AI systems.

The implementation of a digital platform for the innovation registration process, based on AI technology, must take into account the potential challenges that may arise. One key challenge is technological complexity, specifically the fact that developing such an AI-based platform requires significant human, financial, and material resources allocated by the government for its construction. Additionally, there is a need for additional resources to process large volumes of data in a short period of time. Another challenge pertains to the potential additional costs associated with both the creation and the maintenance of such an AI-powered platform. Moreover, specialized personnel are needed to design, manage, and update the system. On the other hand, the implementation of this platform, both in Romania and across other European countries or globally, must comply with legal regulations and standards. Artificial intelligence in such a platform would be used to manage sensitive data, which could pose legal challenges regarding confidentiality and data security. This may require legislative updates at the national level, including in Romania, to allow for the use of AI in managing this type of information.

A further challenge concerns the level of acceptability and trust among researchers regarding the use of such a platform. The adoption of an AI-based system may lead to concerns or resistance from researchers who fear that their data might be compromised or that the information about their innovations could be stolen and used for other purposes. Finally, a challenge related to the creation and implementation of such a platform is the need for its integration with existing systems within innovation registration institutions. This could result in potential barriers to implementation, especially regarding compatibility with existing databases at institutions responsible for registering innovations. All of these challenges must be carefully considered when making the decision to create and implement an AI-based platform to facilitate the innovation registration process within a given country.

This paper advocates for a global, interconnected system of intellectual property protection that would facilitate the mutual recognition of patents and IP rights across borders. Such a system would remove current barriers, reduce the burden on researchers, and foster a more efficient international transfer of technology. The harmonization of IP laws should be seen not only as a priority but also as an opportunity to create a sustainable, innovation-driven global economy. The continued collaboration between nations to streamline IP processes will pave the way for a more knowledge-based and economically competitive global landscape.

5. Conclusions

Research, development, and innovation are fundamental pillars for sustainable economic growth, playing a crucial role in enhancing societal well-being. As developing countries, including Romania, strive to strengthen their innovation ecosystems, the protection of intellectual property (IP) emerges as a key factor in ensuring the commercialization and long-term viability of technological advancements. However, the current registration process presents significant challenges, including administrative burdens, documentation complexities, and legislative inconsistencies, that hinder efficiency and accessibility.

The protection of innovations grounded in disruptive technologies poses substantial challenges, largely due to the rapid rate at which such technologies become obsolete, often within a year, rendering conventional forms of intellectual property protection, such as standard patents, insufficient. This reality underscores the urgent need to align intellectual property legislation with the accelerated pace of technological advancement. In this regard, at the global, European, and national levels, it is essential to rethink and reform existing legal frameworks to ensure that they are adapted to the specific requirements of digital technologies. The development and implementation of such legislation can foster increased innovation, facilitate technology transfer, and contribute to the long-term sustainable development of national economies.

A major barrier to effective IP protection remains the lack of global harmonization in intellectual property regulations. Aligning national policies with international standards would not only simplify procedures but also encourage more researchers and organizations to secure their innovations. In Romania, prolonged registration timelines, excessive documentation requirements, and the absence of a digitalized system for submission and management further complicate the process. These inefficiencies discourage researchers from protecting their intellectual property, ultimately limiting the country's potential for innovation-driven sustainable development. To overcome these challenges, this study highlights the need for a robust legislative framework complemented by advanced digital technologies. The development of an AI-powered online platform for intellectual property registration could significantly enhance efficiency, transparency, and accessibility, enabling researchers to protect their innovations with greater ease. Such a digital transformation would foster a more dynamic innovation landscape, supporting sustainable economic growth and reinforcing Romania's competitiveness in the global market.

For governments and policymakers, it is critical to prioritize the establishment of clear, streamlined IP registration processes and invest in the digitalization of IP management systems. Policy reforms that simplify administrative procedures, reduce documentation requirements, and incorporate artificial intelligence into the registration process can substantially improve the efficiency and attractiveness of intellectual property protection. Furthermore, international cooperation should be encouraged to ensure the alignment of national policies with global standards, facilitating cross-border innovation and fostering a more interconnected and competitive global market.

By addressing these structural barriers and embracing digital solutions, intellectual property management can be transformed into a streamlined and innovation-friendly process. This research underscores the importance of harmonizing regulations, leveraging cutting-edge technologies, and fostering an environment conducive to sustainable technological advancement. Through these efforts, policymakers can create a more supportive environment for innovation, contributing to long-term economic development and global competitiveness.

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References

1. Gaikwad, A.H. A Study of Intellectual Property Rights and Its Significance for Business. *J. Inform. Comput. Sci.* **2020**, *10*, 552–561.
2. Shaikh, I.A.; Randhawa, K. Industrial R&D and National Innovation Policy: An Institutional Reappraisal of the US National Innovation System. *Ind. Corp. Change* **2022**, *31*, 1152–1176.
3. Savitskaya, I.; Podmetina, D. Environmental Influences on Open Innovation: Evidence from Russia. *Int. J. Bus. Excell.* **2013**, *6*, 310–330.
4. World Intellectual Property Organization. What Is Intellectual Property? Available online: <https://www.wipo.int/about-ip/en/> (accessed on 11 December 2024).
5. Kantaros, A. Intellectual Property Challenges in the Age of 3D Printing: Navigating the Digital Copycat Dilemma. *Appl. Sci.* **2024**, *14*, 11448. [CrossRef]
6. Kim, Y.; Sungjin, Y. Global Financial Investment Connectedness, ICT, and Intellectual Property Strategies: A Country-Level Empirical Analysis. *Sustainability* **2024**, *16*, 3282. [CrossRef]
7. Anderson, J.E. Indigenous Knowledge and Intellectual Property Rights. In *International Encyclopedia of the Social & Behavioral Sciences*, 2nd ed.; Elsevier: Orlando, FL, USA, 2015.
8. Singh, R.D. *Law Relating to Intellectual Property*, 1st ed.; Universal Law Publishing Co. Pvt. Ltd.: New Delhi, India, 2004.
9. Kulkarni, A.; Pearce, J.M. Patent Parasites: Non-Inventors Patenting Existing Open-Source Inventions in the 3-D Printing Technology Space. *Inventions* **2023**, *8*, 141. [CrossRef]
10. Saha, C.N.; Bhattacharya, S. Intellectual Property Rights: An Overview and Implications in Pharmaceutical Industry. *J. Adv. Pharm. Technol. Res.* **2011**, *2*, 88–93.
11. Ghafele, R.; Gibert, B. A Counterfactual Impact Analysis of Fair Use Policy on Copyright Related Industries in Singapore. *Laws* **2014**, *3*, 327–352. [CrossRef]
12. Zhu, Y.; Sun, M. The Enabling Effect of Intellectual Property Strategy on Total Factor Productivity of Enterprises: Evidence from China's Intellectual Property Model Cities. *Sustainability* **2023**, *15*, 549.
13. Roffe, P.; Santa Cruz, M. *Intellectual Property Rights and Sustainable Development: A Survey of Major Issues*; United Nations Publication; United Nations: Santiago, Chile, 2007.
14. Bannerman, S. The World Intellectual Property Organization and the Sustainable Development Agenda. *Futures* **2020**, *122*, 102586.
15. Abdel-Latif, A.; Roffe, P. The Interface Between Intellectual Property and Sustainable Development. In *Handbook of Intellectual Property Research: Lenses, Methods, and Perspectives*; Calboli, I., Montagnani, M.L., Eds.; Oxford Academic: Oxford, UK, 2021.
16. Mao, K.; Failler, P. Does Stronger Protection of Intellectual Property Improve Sustainable Development? Evidence from City Data in China. *Sustainability* **2022**, *14*, 14369. [CrossRef]
17. De Miguel Asensio, P.A. *Intellectual Property in European Law*; EPrints Complutense: Bruxelles, Belgium, 2012.
18. Lin, S.H.; Wu, L. Intellectual property rights and law enforcement in developing countries. *Econ. Res.* **2021**, *35*, 143–157. [CrossRef]
19. Nemlioglu, I. A Comparative Analysis of Intellectual Property Rights: A case of Developed versus Developing Countries. *Procedia Comput. Sci.* **2019**, *158*, 988–998.
20. Forero-Pineda, C. The impact of stronger intellectual property rights on science and technology in developing countries. *Res. Policy* **2006**, *35*, 808–824.

21. Auriol, E.; Biancini, S.; Paillacar, R. Intellectual property rights protection and trade: An empirical analysis. *World Dev.* **2023**, *162*, 106072.
22. Bascavusoglu, E.; Zniga, N.P. The Effects of Intellectual Property Protection on International Knowledge Contracting. 2005. Available online: https://www.researchgate.net/publication/23530518_The_effects_of_intellectual_property_protection_on_international_knowledge_contracting (accessed on 17 March 2025).
23. Harhoff, D. Intellectual Property Rights in Europe—Where Do We Stand and Where Should We Go? Prime Minister’s Office—Economic Council of Finland: Helsinki, Finland, 2006; pp. 1–21.
24. Global Information Society Watch. World Intellectual Property Organisation (WIPO). Available online: <https://www.giswatch.org/institutional-overview/civil-society-participation/world-intellectual-property-organisation-wipo> (accessed on 5 December 2024).
25. Maciejewski, M. Proprietatea Intelectuală, Industrială și Comercială. Available online: <https://www.europarl.europa.eu/factsheets/ro/sheet/36/proprietatea-intelectuala-industrial-a-si-comerciala> (accessed on 15 December 2024).
26. State Office for Inventions and Trademarks. Multilateral Agreements and Treaties in the Field of Patents, to Which Romania Is a Party. Available online: <https://osim.ro/en/basic-information-patents/legislation-in-the-ip-field-patents> (accessed on 5 December 2024).
27. Urbinati, A.; Chiaroni, D.; Chiesa, V.; Frattini, F. The role of digital technologies in open innovation processes: An exploratory multiple case study analysis. *RD Manag.* **2018**, *50*, 136–160.
28. Morton, J.; Amrollahi, A.; Wilson, A.D. Digital strategizing: An assessing review, definition, and research agenda. *J. Strateg. Inf. Syst.* **2022**, *31*, 101720.
29. Liu, Q.; Trevisan, A.H.; Yang, M.; Mascarenhas, J. A framework of digital technologies for the circular economy: Digital functions and mechanisms. *Bus. Strategy Environ.* **2022**, *31*, 2171–2192.
30. Quach, S.; Thaichon, P.; Martin, K.D.; Weaven, S.; Palmatier, R.W. Digital technologies: Tensions in privacy and data. *J. Acad. Mark. Sci.* **2022**, *50*, 1299–1323.
31. Kivimaa, P.; Laakso, S.; Lonkila, A.; Kaljonen, M. Moving beyond disruptive innovation: A review of disruption in sustainability transitions. *Environ. Innov. Soc. Transit.* **2020**, *38*, 110–126.
32. Choi, T.; Kumar, S.; Yue, X.; Chan, H. Disruptive Technologies and operations management in the industry 4.0 era and beyond. *Prod. Oper. Manag.* **2021**, *31*, 9–31. [\[CrossRef\]](#)
33. Xu, X.; Lu, Y.; Vogel-Heuser, B.; Wang, L. Industry 4.0 and Industry 5.0—Inception, conception and perception. *J. Manuf. Syst.* **2021**, *61*, 530–535. [\[CrossRef\]](#)
34. Mentsiev, A.U.; Engel, M.V.; Tsamaev, A.M.; Abubakarov, M.V.; Yushaeva, R.S. The concept of digitalization and its impact on the modern economy. In Proceedings of the International Scientific Conference “Far East Con” (ISCFEC 2020), Vladivostok, Russia, 1–4 October 2019; Advances in Economics, Business and Management Research. Atlantis Press: Dordrecht, The Netherlands, 2020. [\[CrossRef\]](#)
35. Caputo, A.; Pizzi, S.; Pellegrini, M.M.; Dabić, M. Digitalization and business models: Where are we going? A science map of the field. *J. Bus. Res.* **2020**, *123*, 489–501. [\[CrossRef\]](#)
36. Gorkhali, A.; Li, L.; Shrestha, A. Blockchain: A literature review. *J. Manag. Anal.* **2020**, *7*, 321–343. [\[CrossRef\]](#)
37. Bodkhe, U.; Tanwar, S.; Parekh, K.; Khanpara, P.; Tyagi, S.; Kumar, N.; Alazab, M. Blockchain for Industry 4.0: A comprehensive review. *IEEE Access* **2020**, *8*, 79764–79800. [\[CrossRef\]](#)
38. Verganti, R.; Vendraminelli, L.; Iansiti, M. Innovation and design in the age of artificial intelligence. *J. Prod. Innov. Manag.* **2020**, *37*, 212–227. [\[CrossRef\]](#)
39. Jan, Z.; Ahamed, F.; Mayer, W.; Patel, N.; Grossmann, G.; Stumptner, M.; Kuusk, A. Artificial intelligence for industry 4.0: Systematic review of applications, challenges, and opportunities. *Expert Syst. Appl.* **2020**, *1194*, 216, 119456. [\[CrossRef\]](#)
40. Hassan, R.; Qamar, F.; Hasan, M.K.; Aman, A.H.M.; Ahmed, A.S. Internet of Things and its applications: A comprehensive survey. *Symmetry* **2020**, *12*, 1674. [\[CrossRef\]](#)
41. Khan, A.A.; Laghari, A.A.; Shaikh, Z.A.; Dacko-Pikiewicz, Z.; Kot, S. Internet of Things (IoT) security with blockchain technology: A state-of-the-art review. *IEEE Access* **2022**, *10*, 122679–122695. [\[CrossRef\]](#)
42. Huang, S.; Wang, B.; Li, X.; Zheng, P.; Mourtzis, D.; Wang, L. Industry 5.0 and Society 5.0—Comparison, complementation and co-evolution. *J. Manuf. Syst.* **2022**, *64*, 424–428. [\[CrossRef\]](#)
43. Khurshid, A.; Huang, Y.; Khan, K.; Cifuentes-Faura, J. Innovation, institutions, and sustainability: Evaluating drivers of household green technology adoption and environmental sustainability of Africa. *Gondwana Res.* **2024**, *132*, 88–102. [\[CrossRef\]](#)
44. Khurshid, A.; Khan, K.; Chen, Y.; Cifuentes-Faura, J. Do green transport and mitigation technologies drive OECD countries to sustainable path? *Transp. Res. Part D Transp. Environ.* **2023**, *118*, 103669. [\[CrossRef\]](#)
45. Khan, K.; Khurshid, A.; Su, C.W. Is Sustainable Innovation an Impetus to Sustainable Development Goals? Evidence From the EU. *Sustain. Dev.* **2024**, 1–6. [\[CrossRef\]](#)

46. Coelho, P.; Bessa, C.; Landeck, J.; Silva, C. Industry 5.0: The arising of a concept. *Procedia Comput. Sci.* **2023**, *217*, 1137–1144. [\[CrossRef\]](#)
47. Singh, T.; Singh, D.; Singh, C.D.; Singh, K. *Industry 5.0*; Wiley Online Library: Hoboken, NJ, USA, 2023; pp. 21–45.
48. Feng, J.; Jaravel, X. Crafting Intellectual Property Rights: Implications for patent assertion entities, litigation, and innovation. *Am. Econ. J. Appl. Econ.* **2020**, *12*, 140–181. [\[CrossRef\]](#)
49. Khouilla, H.; Bastidon, C. Does increased intellectual property rights protection foster innovation in developing countries? A literature review of innovation and catch-up. *J. Int. Dev.* **2023**, *36*, 1170–1188. [\[CrossRef\]](#)
50. Omorov, R.O. Intellectual property and artificial intelligence. *E-Management* **2020**, *3*, 43–49. [\[CrossRef\]](#)
51. Bigliardi, B.; Ferraro, G.; Filippelli, S.; Galati, F. The past, present and future of open innovation. *Eur. J. Innov. Manag.* **2020**, *24*, 1130–1161.
52. Ansell, C.; Miura, S. Can the power of platforms be harnessed for governance? *Public Adm.* **2019**, *98*, 261–276. [\[CrossRef\]](#)
53. Gawer, A. Digital platforms and ecosystems: Remarks on the dominant organizational forms of the digital age. *Innovation* **2021**, *24*, 110–124. [\[CrossRef\]](#)
54. Wegner, D.; Da Silveira, A.B.; Marconatto, D.; Mitrega, M. A systematic review of collaborative digital platforms: Structuring the domain and research agenda. *Rev. Manag. Sci.* **2023**, *18*, 2663–2695.
55. Ramizo, G. Platform playbook: A typology of consumer strategies against algorithmic control in digital platforms. *Inf. Commun. Soc.* **2021**, *25*, 1849–1864. [\[CrossRef\]](#)
56. Giest, S.; Samuels, A. Administrative burden in digital public service delivery: The social infrastructure of library programs for eature review. *Rev. Policy Res.* **2022**, *40*, 626–645.
57. Henman, P. Improving public services using artificial intelligence: Possibilities, pitfalls, governance. *Asia Pac. J. Public Adm.* **2020**, *42*, 209–221.
58. Chen, L.; Tong, T.W.; Tang, S.; Han, N. Governance and design of digital platforms: A review and future research directions on a meta-organization. *J. Manag.* **2021**, *48*, 147–184.
59. Hein, A.; Schreieck, M.; Riasanow, T.; Setzke, D.S.; Wiesche, M.; Böhm, M.; Krcmar, H. Digital platform ecosystems. *Electron. Mark.* **2019**, *30*, 87–98.
60. Sturgeon, T.J. Upgrading strategies for the digital economy. *Glob. Strateg. J.* **2019**, *11*, 34–57.
61. Chen, Y.; Richter, J.I.; Patel, P.C. Decentralized governance of digital platforms. *J. Manag.* **2020**, *47*, 1305–1337.
62. DeLone, W.K.; McLean, E.R. The DeLone and McLean Model of Information Systems Success: A Ten-Year Update. *J. Manag. Inf. Syst.* **2003**, *19*, 9–30.
63. Bolton, R.N.; Drew, J.H. A Multistage Model of Customers' Assessments of Service Quality and Value. *J. Consum. Res.* **1991**, *17*, 375–384.
64. Parasuraman, A.; Zeithaml, V.A.; Berry, L.L. A Conceptual Model of Service Quality and Its Implications for Future Research. *J. Mark.* **1985**, *49*, 41–50. [\[CrossRef\]](#)
65. Baker, J.; Parasuraman, A.; Grewal, D.; Voss, G.B. The Influence of Multiple Store Environment Cues on Perceived Merchandise Value and Patronage Intentions. *J. Mark.* **2002**, *66*, 120–141. [\[CrossRef\]](#)
66. Parasuraman, A.; Zeithaml, V.A.; Berry, L.L. SERVQUAL: A Multiple-Item Scale for Measuring Consumer Perceptions of Service Quality. *J. Retail.* **1988**, *64*, 12–40.
67. Dodds, W.B.; Monroe, K.B.; Grewal, D. Effects of Price, Brand, and Store Information on Buyers' Product Evaluations. *J. Mark. Res.* **1991**, *28*, 307–319.
68. Agarwal, S.; Teas, K.R. Perceived Value: Mediating Role of Perceived Risk. *J. Mark. Theory Pract.* **2001**, *9*, 1–14.
69. Stancu, D. *Transferul Tehnologic și Proprietatea Intelectuală în România*; Romanian Academy, The School of Advanced Studies of the Romanian Academy, Doctoral School of Economic Sciences, Institute of National Research in Economy “Costin C. Kirițescu”: Bucharest, Romania, 2024.
70. Rimmer, M. Back to the Future: The Digital Millennium Copyright Act and the Trans-Pacific Partnership. *Laws* **2017**, *6*, 11. [\[CrossRef\]](#)
71. Chen, Q.; Xia, X.; Hui, Y.; Zhang, S. Urban Intellectual Property Strategy and University Innovation: A Quasi-Natural Experiment Based on the Intellectual Property Model City of China. *Systems* **2024**, *12*, 21. [\[CrossRef\]](#)
72. Li, W.; Li, B. Intellectual Property Pledge Financing and Enterprise Innovation: Based on the Perspective of Signal Incentive. *Sustainability* **2023**, *15*, 10448. [\[CrossRef\]](#)
73. Markman, G.D.; Gianiodis, P.T.; Phan, P.H.; Balkin, D.B. Innovation Speed: Transferring University Technology to Market. *Res. Policy* **2005**, *34*, 1058–1075.
74. Cao, Y.; Elahi, E.; Khalid, Z.; Li, P.; Sun, P. How Do Intellectual Property Rights Affect Green Technological Innovation? Empirical Evidence from China. *Sustainability* **2023**, *15*, 7762. [\[CrossRef\]](#)
75. Rimmer, M. Beyond the Paris Agreement: Intellectual Property, Innovation Policy, and Climate Justice. *Laws* **2019**, *8*, 7. [\[CrossRef\]](#)
76. Tașcu, A.V.; Radu, A.V.; Stoica, I.; Dobrescu, A.I. Online Decision Purchase Process of Medicines. *Farmacia* **2017**, *65*, 19–22.

77. Niculescu, M.; Honțaru, O.; Popescu, G.; Sterian, A.G.; Dobra, M. Challenges of Integrating New Technologies for Orthopedic Doctors to Face up to Difficulties During the Pandemic Era. *Health Care* **2023**, *11*, 1524. [[CrossRef](#)] [[PubMed](#)]
78. Mustață, M.A.; Răpan, I.; Dumitrescu, L.; Dobrevă, H.; Dimov, P.; Lis, A.; Révayová, E.; Marineneanu, V.; Buluc, R.; Olariu, C.; et al. Assessing the Truthfulness of Security and Defence News in Central and Eastern Europe: The Role of Cognitive Style and the Promise of Epistemic Sophistication. *Appl. Cogn. Psychol.* **2023**, *37*, 1384–1396.
79. Răpan, I.; Zaman, G.; Suci, M.C.; Purcărea, V.L.; Jude, C.R.; Radu, A.V.; Catană, A.; Radu, A.C. A Better Integration of Industrial Robots in Romanian Enterprises and the Labour Market. *Appl. Sci.* **2022**, *12*, 6014. [[CrossRef](#)]
80. Xia, X.; Huang, T.; Zhang, S. The Impact of Intellectual Property Rights City Policy on Firm Green Innovation: A Quasi-Natural Experiment Based on a Staggered DID Model. *Systems* **2023**, *11*, 209. [[CrossRef](#)]
81. Hamza, R.; Pradana, H. A Survey of Intellectual Property Rights Protection in Big Data Applications. *Algorithms* **2022**, *15*, 418. [[CrossRef](#)]
82. Sarnoff, J. *Research Handbook on Intellectual Property and Climate Change*; Edward Elgar: Cheltenham/Northampton, UK, 2016.
83. Sun, H.; Liu, J.; Chen, B.; Yang, L. Exploring Intellectual Property in the Digital Realm: A Bibliometric Study on Research on the Management and Protection of Data-Based Intellectual Property. *Information* **2024**, *15*, 780. [[CrossRef](#)]
84. Unnikrishnan, A. Analyzing the impact of emerging technologies on Intellectual property rights (IPR): A comprehensive study on the challenges and opportunities in the digital age. *Law World* **2024**, *10*, 66–79.
85. Adams, R. The Evolution of Intellectual Property Rights in the Digital Age. *J. Mod. Law Policy* **2023**, *3*, 52–63.
86. Hilgartner, S. Intellectual Property and the Politics of Emerging Technology: Inventors, Citizens, and Powers to Shape the Future. *Chic.-Kent Law Rev.* **2008**, *84*, 197–224.
87. Kongolo, T. *Intellectual Property and Emerging Technologies Generated Global IP Issues and Challenges*, 1st ed.; Routledge: London, UK, 2024.

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